LITERATURE SURVEY

SignVision – Real-Time Sign Language Interpreter

1. Project Overview

SignVision explores sign-language hand gesture recognition using machine learning and deep learning. The project compares Logistic Regression, Random Forest and CNN, then displays the predicted sign through a user-friendly interface. The papers below were selected because they study sign-language recognition, visual features, classifiers or real-time systems.

2. Research Papers and Summaries

1. American Sign Language Alphabet Recognition Using Convolutional Neural Networks with Multiview Augmentation and Inference Fusion

Authors / year: Wenjin Tao, Ming C. Leu and Zhaozheng Yin (2018)

Method: The authors used CNNs with depth images from a Microsoft Kinect camera. They generated multiple views of training images and combined predictions from different views.

Main finding: Multiview augmentation and prediction fusion improved ASL alphabet recognition on two public datasets.

Limitation / relevance: The approach used depth-camera images, which may not be available in a simple webcam-based project.

Research link: https://doi.org/10.1016/j.engappai.2018.09.006

2. Sign Language Recognition: A Deep Survey

Authors / year: Razieh Rastgoo, Kourosh Kiani and Sergio Escalera (2021)

Method: This review organizes deep-learning research for isolated and continuous sign recognition, and discusses datasets, evaluation metrics and model complexity.

Main finding: It reports progress from deep learning while highlighting challenges caused by signer variation, motion and continuous signing.

Limitation / relevance: A survey does not itself provide one ready-to-use model; it identifies broader research challenges.

Research link: https://doi.org/10.1016/j.eswa.2020.113794

3. Machine Learning Methods for Sign Language Recognition: A Critical Review and Analysis

Authors / year: I. A. Adeyanju, O. O. Bello and M. A. Adegboye (2021)

Method: The authors reviewed research on intelligent sign-recognition systems, including image segmentation, feature extraction and classification methods.

Main finding: The review shows that machine learning methods are widely studied, but reliable recognition remains an open problem.

Limitation / relevance: The paper reviews many studies rather than testing the exact three models used in SignVision.

Research link: https://doi.org/10.1016/j.iswa.2021.200056

4. A Hybrid Approach for Bangla Sign Language Recognition Using Deep Transfer Learning Model with Random Forest Classifier

Authors / year: Research article (2022), Expert Systems with Applications

Method: The study combines CNN-based transfer learning with a Random Forest classifier and evaluates Bangla alphabet and numeral datasets. It also uses background removal.

Main finding: The hybrid method was evaluated using accuracy, precision, recall and F1-score; reported performance differed between character and digit recognition.

Limitation / relevance: Results are specific to the Bangla sign datasets and may not transfer directly to another sign language or dataset.

Research link: https://www.sciencedirect.com/science/article/pii/S0957417422019327

5. Sign Language Recognition Based on Dual-Path Background Erasure Convolutional Neural Network

Authors / year: Junming Zhang and colleagues (2024)

Method: The authors proposed a CNN approach designed to reduce background influence during sign-language image recognition.

Main finding: The work focuses on separating useful hand-sign information from distracting background details.

Limitation / relevance: A deep model may require more training and computation than simpler classifiers, and performance depends on the dataset and conditions.

Research link: https://www.nature.com/articles/s41598-024-62008-z

6. A Machine Learning-Driven Web Application for Sign Language Learning

Authors / year: Research article (2024), Frontiers in Artificial Intelligence

Method: The paper discusses a web-based sign-language learning application and reviews recognition methods including CNNs and traditional classifiers.

Main finding: It emphasizes the value of presenting recognition or learning functions through an accessible web application.

Limitation / relevance: The supported signs and results depend on the dataset and system scope; learning tools do not necessarily translate continuous conversation.

Research link:
https://www.frontiersin.org/journals/artificial-intelligence/articles/10.3389/frai.2024.1297347/full

3. Comparison with SignVision

The papers cover CNN-based image recognition, deep-learning surveys, traditional machine-learning methods, hybrid CNN–Random Forest approaches and web-based applications. This provides background for comparing Logistic Regression, Random Forest and CNN in SignVision. The literature does not prove which model will perform best on the project dataset; that must be determined from the project's own evaluation results.

4. Research Gap

The reviewed research identifies continuing challenges such as background and lighting variation, differences between signers, dataset limitations and the computational cost of some deep models. SignVision addresses a project-level need by comparing three selected models on its chosen dataset and showing predictions in a simple interface. Its actual scope should be described accurately—for example, alphabet or isolated gestures if that is what the implementation recognizes.

5. Conclusion

Previous studies show that CNNs can learn visual patterns from sign images, while traditional classifiers and hybrid models are also useful. Reviews highlight that accuracy alone is not enough: dataset diversity and real-world conditions matter. These findings support evaluating several models and reporting measured results for SignVision.

6. References

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